Summer 2025
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Introduction to Natural Language Processing

Worksheet for the Project

Philipps Universität Marburg

The project consists of four deliverables, three of which will be graded:

1. Team formation and track selection (not graded)
2. Project proposal (20 pts)
3. Technical solution (50 pts)
4. Project report (30 pts)

The project is based on the data of the 2025 SemEval Shared Task 11 (Muhammad et al. 2025). You can choose to work on **one** of two tracks (you can find the training data sets for both tracks on Ilias.):

- **Track A (Multi-label Emotion Detection)**: Given a target text snippet, predict the perceived emotion(s) of the speaker. Specifically, select whether each of the following emotions apply: joy, sadness, fear, anger, surprise, or disgust. In other words, label the text snippet with: joy (1) or no joy (0), sadness (1) or no sadness (0), anger (1) or no anger (0), surprise (1) or no surprise (0), and disgust (1) or no disgust (0).
- **Track B (Emotion Intensity)**: Given a target text and a target perceived emotion, predict the intensity for each of the classes. The set of the perceived emotions includes: joy, sadness, fear, anger, surprise, or disgust. The set of ordinal intensity classes includes: 0 for no emotion, 1 for a low degree of emotion, 2 for a moderate degree of emotion, and 3 for a high degree of emotion.

1 Milestone 1

For the first milestone, form a team of 4 or 5 members (numbers fixed) and select which Track you want to work on. You can use the Matrix chat or the tutorial session to find team mates. It is not necessary that all members are part of the same tutorial group, although it might be helpful. You will continue to work in this group for all milestones of the project. To register your choise, submit a file with the names of the team members and the track you selected on Ilias.
Deadline: 23.05.2025 23:59:59

2 Milestone 2

For your chosen track, **write a task description**. Use the templates provided on Ilias (LaTeX or Word). Write no more than four pages describing:

- the task and your motivation for choosing the task
- the dataset
- the methodology you are planning to use
- the way in which you are planning to evaluate your results

You can achieve a maximum of 20 points in this task. Your submission will be assessed based on the following rubric:

Criterion	Insufficient	Sufficient	Good	Excellent
Task Description	No or wrong task description (0pt)	Broad but accurate task description (2.5pt)		Accurate task description including all relevant aspects (5 pt)
Motivation	No (convincing) motivation provided (0pt)			Convincing motivation provided (1pt)
Dataset	Important aspects of the dataset are not described (0pt)	The most important aspects of the dataset with regard to the task are described (5pt)	All aspects of the dataset that are relevant to the task are describe and visualisations are provided where useful (7.5pt)	In addition to the criteria from "Good" first analyses that go beyond pure descriptive analysis has been conducted on the dataset. (10pt)
Methodology	No methodology is describe on how to approach to ask. (0 pt)	A methodology that is purely based on methods introduced in the lecture and is appropriate to tackle the problem is described (1pt)		An appropriate methodology that goes beyond methods introduced in the lectures is described (2pt)
Evaluation	No (convincing) evaluation strategy is described (0pt)	One convincing evaluation metric or approach is described (1pt)		More than one convingin evaluation metric or approach is described (2pt)

Please keep in mind that you need to achieve at least 50% of the total points in the project in order to be admitted to the exam. You have to submit the milestone on Ilias with the group you registered. The submission deadline is a hard deadline, failure to submit in time will results in 0 points.

Deadline: 30.05.2025 23:59:59

3 Milestone 3

Implement a Python script that can perform the classification task of your chosen track. You are not restricted to technologies introduced in the lecture, you can train, fine-tune, or use any model or algorithm (but your choice will have influence on the assessment, see rubric below).

Your final submission needs to contain a file main.py which has a function `predict` which takes as argument the location of a `.csv` file that is structured in the same way as your training data and returns an array of predictions for each item in the `.csv` file (i.e. an array of arrays indicating [anger, fear, you, sadness, surprise]). This functions needs to be *executable* during the assessment of your project, i.e. if you rely on external APIs your code needs to include a working API key. If your code cannot be executed, you will receive zero points for "Classification Capabilities". Submit your code as .zip file on Ilias.

You can achieve a maximum of 50 points in this task. Your submission will be assessed based on the following rubric:

Criterion	Insufficient	Sufficient	Good
Documentation	The code is not documented (0 pt.)	Basic documentation is included and useful names have been chosen for variables and functions (5 pts.)	The code is well documented and everything is named consistently and appropriately (10 pts.)
Classification Capabilities	Your approaches achieves less than 50% accuracy of the average accuracy of the top 3 approaches. (0 pt.)	Your approaches achieves 50% accuracy of the average accuracy of the top 3 approaches. (10 pts.)	Your approach achieves more than 50% accuracy of the average accuracy of the top 3 approaches. Your points will be: $max(20, \frac{accuracy}{avg(top3)} * 20)$
Technical Quality	The overall technical quality of the project (efficiency, effectiveness, correctness, maintainability, ...) is poor (0 pt.)	The overall technical quality (efficiency, effectiveness, correctness, maintainability, ...) of the project is sufficient (5 pts.)	The overall technical quality (efficiency, effectiveness, correctness, maintainability, ...) of the project is good (10 pts.)
Technical Difficulty	The difficulty of the implemented solution is trivial (e.g. solely relies on external APIs) (0 pt.)	The difficulty of the implemented solution is basic, it is based on simple approaches introduced in the lecture and tutorials (5 pts.)	The difficulty of the implemented solution is high, it also implements non-trivial techniques (10 pts.)

Deadline 04.07.2025 23:59:59.

4 Milestone 4

Write a final **project report**. Use the templates provided on Ilias (LaTeX or Word). Write no more than six pages containing:

- an introduction summarising the data and your motivation for choosing it
- your technical solution (what techniques did you use and how did you implement them)
- evaluation results on the provided test data
- reflection on how your final project differs from the original plan you made for Milestone 2

You can achieve a maximum of 30 points in this task. Your submission will be assessed based on the following rubric:

Criterion	Insufficient	Sufficient	Good
Introduction	No motivation provided and/or data not appropriately described(0 pts.)	Data described appropriately but no conniving motivation provided (2.5 pts.)	Good description of the data and convining motivation provided (5 pts.)
Technical Solution	Technical solution inappropriately described (0 pts.)	Technical description is brief but accurate (5 pts.)	Good technical description, references used where appropriate, visualisations used where appropriate (10 pts.)
Evaluation	No (relevant) evaluation described. (0 pts.)	Basic evaluation with at least relevant metrics is described. (2.5 pts.)	Detailed evaluation consisting of metrics and qualitative analysis of errors. (5 pts.)
Reflection	No (accurate) reflection provided (0 pts.)	Reflection is provided but incomplete, shallow or uncritical (2.5 pts.)	Critical and complete reflection provided that elaborates and justifies deviations from the original plan (5 pts.)
Document	Document is not formatted according to the provided template or is lacking a clear structure. (0 pts.)	The document has a basic structure, follows the template and is overall of good quality. (2.5 pts.)	CThe report is very well written, nicely structured, and uses tables, images, and similar elements where appropriate. (5 pts.)

Deadline 18.07.2025 23:59:59.

References

Muhammad, Shamsuddeen Hassan et al. (2025). "SemEval Task 11: Bridging the Gap in Text-Based Emotion Detection". In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Vienna, Austria: Association for Computational Linguistics.